

IoT Concepts & Architecture Study

CodeOrbit Tech – Internet of Things (IoT) Internship

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Task	Task 1 – IoT Concepts & Architecture Study
Topic	Basic Architecture of an IoT System
Submission	PDF Report

1. Introduction to Internet of Things (IoT)

The Internet of Things (IoT) is a technology in which physical devices are connected to a network so that they can collect, exchange, process, and use data. IoT devices can contain sensors, controllers, communication modules, and software. The collected data can be sent to a cloud or local server for storage and analysis and can then be presented to users through an application or dashboard.

Examples of IoT include smart homes, smart agriculture, health monitoring systems, industrial monitoring, smart cities, and connected vehicles.

2. Objective

The objective of this study is to understand the basic layers of an IoT system and how data moves from a physical sensor to a user application.

3. Basic IoT Architecture

APPLICATION LAYER	Mobile app / Web dashboard Displays data and alerts to the user
CLOUD / PROCESSING LAYER	Cloud server / Database / Analytics Stores and processes sensor data
CONNECTIVITY LAYER	Wi-Fi / Bluetooth / LoRa / Cellular Transfers data between devices and servers
SENSING / DEVICE LAYER	Temperature / Humidity / Motion sensors Collects information from the physical world

Data flow: Physical environment → Sensor → Connectivity → Cloud/Processing → Application → User.

4. Explanation of the IoT Layers

4.1 Sensing / Device Layer

This is the physical layer of an IoT system. Sensors measure physical conditions such as temperature, humidity, light, pressure, distance, motion, or gas concentration. A microcontroller such as an ESP32 or Arduino can read the sensor values. Actuators can also be used to perform actions, such as switching a motor, relay, light, or fan.

Real-world example: A temperature sensor in a smart room measures the room temperature.

4.2 Connectivity Layer

The connectivity layer transfers data from IoT devices to another device, gateway, or cloud service. Common communication technologies include Wi-Fi, Bluetooth, Zigbee, LoRa/LoRaWAN, Ethernet, and cellular networks. The choice depends on range, power consumption, data rate, and application requirements.

Real-world example: An ESP32 sends temperature readings to a server using Wi-Fi.

4.3 Cloud / Processing Layer

The cloud or processing layer receives IoT data, stores it, processes it, and can perform analytics. It may also manage devices and generate alerts. Databases are used to keep historical sensor readings so that trends can be analyzed.

Real-world example: A cloud platform stores temperature readings and detects when the temperature exceeds a limit.

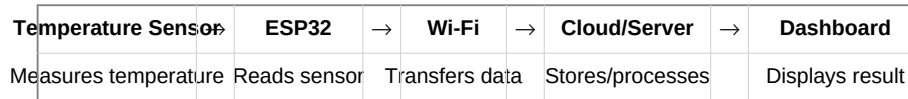
4.4 Application Layer

The application layer presents useful IoT information to users through a mobile application, web application, dashboard, or monitoring system. It may show live readings, historical graphs, device status, and alerts. Users can sometimes send commands back to devices.

Real-world example: A smart-home dashboard displays room temperature and allows the user to control a fan.

5. Real-World Example – Smart Home Temperature Monitoring

Consider a smart-home room-temperature monitoring system. A temperature sensor continuously measures the room temperature. An ESP32 reads the sensor value and sends it over Wi-Fi. A cloud or local server stores the readings. A web or mobile dashboard displays the current temperature and historical graph. If the temperature becomes too high, the system can generate an alert or activate a fan through a relay.



6. Benefits of IoT

- Real-time monitoring of physical conditions.
- Remote access and control of connected devices.
- Automation of repetitive tasks.
- Better data collection and decision-making.
- Predictive maintenance and early fault detection.
- Improved efficiency in homes, industries, agriculture, and cities.

7. Challenges of IoT

- Security and privacy risks.
- Dependence on reliable connectivity.
- Device power and battery limitations.
- Large amounts of data to store and process.
- Compatibility between different devices and platforms.

8. Conclusion

IoT connects physical objects to digital systems so that information can be collected, transmitted, processed, and presented to users. The basic architecture can be understood through four main parts: the sensing/device layer, connectivity layer, cloud/processing layer, and application layer. Together, these layers enable real-time monitoring, automation, remote control, and data-driven decisions.

9. References

- General IoT architecture and concepts from standard IoT learning resources.
- IBM – Internet of Things (IoT) concepts.
- AWS – IoT concepts and cloud-based IoT architecture.